

NECKLACES, THE LETTER SWAP, AND ZAGIER-REDUCED FORMS OF DISCRIMINANT $N^2 - 4$

ABSTRACT. Let $\mathcal{M}$ be the free monoid generated by $L = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $R = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$, and let $\Phi(N) = \#\{X \in \mathcal{M} : \text{tr } X = N\}$ be the state count of the Farey fraction spin chain of Kleban and Özlük. Let $Z(D)$ denote the number of Zagier-reduced integral binary quadratic forms of discriminant D . The Online Encyclopedia of Integer Sequences has recorded since 2015 the conjecture that $\Phi(N) = 2Z(N^2 - 4)$ (A264598, A264597, A257007), with the annotation “It would be nice to have a proof”. We prove it, in three ways.

First, an explicit divisor parametrisation of the Zagier-reduced forms of a non-square discriminant D (Lemma 3.1) identifies $Z(D)$ with the quantity $\Upsilon^<(D)$ appearing as an intermediate step in Technau’s 2023 analysis of Φ ; the conjecture is then immediate from his equation (3.2). Second, the same parametrisation combined with a fixed-point-free involution on divisor pairs yields a short self-contained proof from first principles, independent of the literature. Third, and structurally, we bend the words of $\mathcal{M}$ into necklaces: cyclic words of trace N are the conjugacy classes of trace N , the length of the Zagier cycle of a class equals the number of R ’s in its word, and conjugation by $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ — a matrix of determinant -1 , hence not a symmetry of $\text{SL}_2(\mathbb{Z})$, but a trace-preserving one — permutes the classes of trace N while exchanging the two letter counts. The factor 2 in the conjecture is exactly the two colours of bead. This third argument proves the finer, per-class statement of which the conjecture is the total.

We then introduce the *cut-sum* $S(w)$ of a necklace, the sum of the matrices of all its cuts: a well-defined matrix-valued invariant strictly finer than the trace. We prove $\det S = n + \sum \text{tr}[u, v]$, summed over pairs of cuts with u, v the two arcs they determine, and identify $(\text{tr } S)^2 - 4 \det S$ as the Minkowski norm of a sum of pole vectors. Finally we record the total-matrix identity $\sum_{\text{tr } X=N} X = \frac{N}{2} \Phi(N) I + (\sum_a \sigma(a(N-a)-1)) J$, whose σ -half appears not to have been studied. All assertions were verified by exact integer computation.

1. INTRODUCTION

1.1. The spin chain. Let

$$L = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad R = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix},$$

and let $\mathcal{M} \subset \text{SL}_2(\mathbb{Z})$ be the monoid they generate. Kleban and Özlük [5] introduced a statistical-mechanical model — the *Farey fraction spin chain* — whose states are the elements of $\mathcal{M}$ and whose energy is the logarithm of the trace. The associated counting function is

$$\Phi(N) = \#\{X \in \mathcal{M} : \text{tr } X = N\}, \quad N \geq 3. \tag{1}$$

It begins 2, 6, 8, 14, 14, 24, 16, 40, 26, 36, $\dots$ and is A264598 in the OEIS [7]. Its behaviour is notoriously irregular: Kleban and Özlük conjectured $\Phi(N) \sim c_0 N \log N$, and Peter [9] showed that no such asymptotic can hold, since $\Phi(N)/(N \log N)$ possesses a smooth limiting distribution rather than a limit. Attention has since concentrated on the summatory function, treated by Kallies–Özlük–Peter–Snyder [4], Boca [1] and Ustinov [12]. Most recently Technau [11] reduced the estimation of Φ to a divisor problem for quadratic polynomials solved by Hooley [3] and, in full generality, by Bykovskii–Ustinov [2].

1.2. Zagier-reduced forms. Independently, Zagier’s reduction theory for indefinite binary quadratic forms [13, §13] calls an integral form $Q = (A, B, C)$, meaning $Ax^2 + Bxy + Cy^2$, of discriminant $D = B^2 - 4AC > 0$ *reduced* when

$$A \geq 1, \quad C \geq 1, \quad B > A + C. \quad (2)$$

Write $Z(D)$ for the number of such forms. The values of $Z(N^2 - 4)$ for $N = 3, 4, 5, \dots$ are 1, 3, 4, 7, 7, 12, 8, 20, 13, 18, $\dots$, the sequence A257007 [8].

1.3. The conjecture. Comparing the two lists, Sloane observed in November 2015 that each term of A264598 is twice the corresponding term of A257007, and recorded in A264598 the remark “*This also appears to be the same as A257007 multiplied by 2. It would be nice to have a proof.*” The same caveat is repeated in A257007, whose MAPLE program carries the warning that it should not be used to extend the sequence until the identity is proved. Both annotations stand today. The present note removes them.

Theorem 1.1. *For every integer $N \geq 3$,*

$$\Phi(N) = 2Z(N^2 - 4).$$

1.4. Three proofs, and what is new in each. We give three arguments, and we are explicit about how much of each is already in the literature.

(i) *A deduction from [11].* Technau’s route to his Theorem 2 passes through the intermediate identity $\Phi(N) = 2\Upsilon^<(N^2 - 4)$ [11, (3.2)], where $\Upsilon^<$ is a certain restricted divisor sum. Our Lemma 3.1 below shows that $\Upsilon^<(D)$ is precisely the number of Zagier-reduced forms of discriminant D ; Theorem 1.1 follows at once. The factor 2 was therefore already proved in 2023. What was missing was the dictionary entry: [11] does not mention Zagier reduction, and the OEIS entries do not mention [11]. The new content of this proof is Lemma 3.1.

(ii) *An elementary self-contained proof* (§4). Combining Lemma 3.1 with a fixed-point-free involution on divisor pairs gives a proof of Theorem 1.1 occupying half a page and using nothing beyond the freeness of $\mathcal{M}$. The involution is a re-derivation, in the coordinates natural for Zagier reduction, of [11, Lemma 4].

(iii) *A structural proof* (§5), which is the reason for this note. Bend the words of $\mathcal{M}$ into loops. Cyclic words — *necklaces* — of trace N are exactly the $\mathrm{SL}_2(\mathbb{Z})$ -conjugacy classes of trace N ; each contributes to $\Phi(N)$ once for each of its cuts, so $\Phi(N) = \sum_C (\#L + \#R)$. On the other side, the length of the Zagier cycle of a class equals $\#R$ alone (Proposition 5.2). The two are reconciled by the involution induced by conjugation with $J = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, which interchanges L and R : it has determinant -1 , so it is not an inner symmetry of $\mathrm{SL}_2(\mathbb{Z})$, but it preserves trace and determinant, hence permutes the classes of trace N among themselves while exchanging the two letter counts. Summed over a set that it permutes, the counts must agree. This proof establishes the finer per-class statement, of which Theorem 1.1 is the total.

1.5. Further results. Section 6 determines the necklaces fixed by the letter swap: they are exactly the words $u\bar{u}$, and their traces are exactly the integers $s^2 + 2$. Section 7 introduces the cut-sum, a matrix-valued necklace invariant finer than the trace, and proves a commutator formula for its determinant together with a Minkowski interpretation of its discriminant. Section 8 records a total-matrix identity whose off-diagonal entry is a sum of $\sigma(a(N - a) - 1)$, a quantity that seems not to have been considered. Section 9 documents the computations, and Section 10 lists what we could not settle.

Conventions. $d(\cdot)$ and $\sigma(\cdot)$ are the number-of-divisors and sum-of-divisors functions. Divisors are positive. Discriminants are positive and never perfect squares; note $N^2 - 4$ is a square only for $N = 2$.

2. THE FREE MONOID AND ITS TRACE FIBRES

Proposition 2.1. $\mathcal{M} \cup \{I\}$ is precisely the set of 2×2 integer matrices with non-negative entries and determinant 1, and it is free on $\{\mathbf{L}, \mathbf{R}\}$.

Proof. Freeness is due to Kleban and Özlük [5]; see also Nathanson [6] for a proof of a more general statement by the ping-pong lemma.

For surjectivity, let $X = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ have non-negative entries and $ad - bc = 1$, and suppose $X \neq I$. We claim that either $a \geq c$ and $b \geq d$, or $c \geq a$ and $d \geq b$. Suppose not; then one of the following holds.

- $a < c$ and $d < b$. All entries being non-negative integers, $c \geq a + 1$ and $b \geq d + 1$, so $bc \geq (a + 1)(d + 1) = ad + a + d + 1 > ad$, contradicting $ad - bc = 1$.
- $c < a$ and $b < d$. Symmetrically $ad \geq (b + 1)(c + 1) = bc + b + c + 1$, so $1 = ad - bc \geq b + c + 1$, forcing $b = c = 0$ and then $ad = 1$, i.e. $X = I$, which was excluded.

(The two remaining combinations, $a < c$ with $c < a$ and $b < d$ with $d < b$, are vacuous.) In the first case $\mathbf{L}^{-1}X = \begin{pmatrix} a-c & b-d \\ c & d \end{pmatrix}$ again has non-negative entries and determinant 1; in the second, $\mathbf{R}^{-1}X = \begin{pmatrix} a & b \\ c-a & d-b \end{pmatrix}$ does. Since the sum of the entries strictly decreases at each such step unless $X = I$, iteration expresses X as a word in $\mathbf{L}$ and $\mathbf{R}$. $\square$

Proposition 2.2. Let $N \geq 3$ and put $\mathcal{N}_a = a(N - a) - 1$. The map $X \mapsto (X_{11}, X_{12})$ is a bijection

$$\{X \in \mathcal{M} : \text{tr } X = N\} \longrightarrow \{(a, b) \in \mathbb{Z}^2 : 1 \leq a \leq N - 1, b \geq 1, b \mid \mathcal{N}_a\}.$$

Consequently

$$\Phi(N) = \sum_{a=1}^{N-1} d(a(N - a) - 1). \quad (3)$$

Proof. Write $X = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with non-negative entries, $a + d = N$, $ad - bc = 1$; then $bc = ad - 1 = \mathcal{N}_a$. If $a = 0$ or $a = N$ then $\mathcal{N}_a = -1 < 0$, impossible for non-negative b, c ; so $1 \leq a \leq N - 1$ and $\mathcal{N}_a \geq (N - 1) - 1 \geq 1$. Hence $b \geq 1$, $b \mid \mathcal{N}_a$, and X is recovered from (a, b) by $d = N - a$, $c = \mathcal{N}_a/b$. Conversely every such pair produces a matrix with non-negative entries, determinant 1 and trace N , which lies in $\mathcal{M}$ by Proposition 2.1 (it is not I since $N \geq 3$). $\square$

Remark 2.3. Identity (3) is Technau's Theorem 2 [11] in different coordinates. Indeed, put $m = N - 2a$; as a runs over $1, \dots, N - 1$, m runs over exactly those integers with $m^2 < N^2 - 4$ and $m^2 \equiv N^2 - 4 \pmod{4}$, and $((N^2 - 4) - m^2)/4 = a(N - a) - 1$. Thus (3) reads $\Phi(N) = \Upsilon(N^2 - 4)$ with $\Upsilon(X) = \sum_{m^2 < X, m^2 \equiv X \pmod{4}} d((X - m^2)/4)$, which is [11, Theorem 2]. We include the short proof above to keep the present note self-contained.

3. A DIVISOR PARAMETRISATION OF THE ZAGIER-REDUCED FORMS

Throughout this section $D > 0$ is an integer, not a perfect square, with $D \equiv 0, 1 \pmod{4}$. Put

$$P(D) = \left\{ (h, k) \in \mathbb{Z}^2 : k^2 < D, k^2 \equiv D \pmod{4}, h \geq 1, h \mid \frac{D - k^2}{4} \right\}, \quad (4)$$

and for $(h, k) \in P(D)$ write $g = g(h, k) = \frac{D - k^2}{4h} \geq 1$ for the complementary divisor, so that $4hg = D - k^2$. Let

$$P^+(D) = \{(h, k) \in P(D) : 2h + k > \sqrt{D}\}. \quad (5)$$

Lemma 3.1. *The assignment $(A, B, C) \mapsto (h, k) := (A, B - 2A)$ is a bijection from the set of Zagier-reduced forms of discriminant D onto $P^+(D)$, with inverse $(h, k) \mapsto (A, B, C) = (h, k + 2h, h + k - g(h, k))$. In particular*

$$Z(D) = \#P^+(D).$$

Proof. Let (A, B, C) satisfy (2) with $B^2 - 4AC = D$, and set $h = A \geq 1$, $k = B - 2A$. Then

$$D = B^2 - 4AC = (k + 2h)^2 - 4hC = k^2 + 4h(h + k - C),$$

so with $g := h + k - C$ we have $4hg = D - k^2$; in particular $k^2 \equiv D \pmod{4}$ and, provided $g \geq 1$, also $h \mid (D - k^2)/4$ and $k^2 < D$. Now

$$B > A + C \iff k + 2h > h + C = h + (h + k - g) \iff g \geq 1,$$

and $g \geq 1$ is equivalent to $k^2 < D$ because $4hg = D - k^2$ and $h \geq 1$. Next, since $h \geq 1$,

$$(2h + k)^2 > D \iff 4h^2 + 4hk + k^2 > 4hg + k^2 \iff h + k > g \iff C = h + k - g \geq 1.$$

If $C \geq 1$ then $h + k > g \geq 1$, so $2h + k = h + (h + k) > 0$ and therefore $2h + k > \sqrt{D}$; conversely $2h + k > \sqrt{D} > 0$ gives $(2h + k)^2 > D$ and hence $C \geq 1$. Thus the constraints (2) correspond exactly to membership in $P^+(D)$. The displayed inverse map is immediate, and both composites are the identity. $\square$

Corollary 3.2. $Z(D) = \Upsilon^<(D)$, where $\Upsilon^<$ is the restricted divisor sum of [11, (3.3)].

Proof. By definition [11, (3.3)], $\Upsilon^<(D) = \sum_{m^2 < D, m^2 \equiv D(4)} d_{(m+\sqrt{D})/2}((D - m^2)/4)$, where $d_n(x) = \#\{k \mid x : k < n\}$; that is, $\Upsilon^<(D)$ counts the pairs (h, m) with $h \mid (D - m^2)/4$ and $2h - m < \sqrt{D}$. Substituting $k = -m$, which permutes the index set, $\Upsilon^<(D) = \#\{(h, k) \in P(D) : 2h + k < \sqrt{D}\} = \#P(D) - \#P^+(D)$; equality is excluded because D is not a square. By [11, Lemma 4], $\Upsilon(D) = \#P(D) = 2\Upsilon^<(D)$ for non-square D , whence $\#P^+(D) = \Upsilon^<(D)$, and Lemma 3.1 concludes. $\square$

First proof of Theorem 1.1. By [11, (3.2)], $\Phi(N) = 2\Upsilon^<(N^2 - 4)$ for $N \geq 3$, and $N^2 - 4$ is not a square. Corollary 3.2 identifies $\Upsilon^<(N^2 - 4)$ with $Z(N^2 - 4)$. $\square$

4. AN ELEMENTARY PROOF

The following is independent of [11] and of all reduction theory; it uses only Proposition 2.2 and Lemma 3.1.

Lemma 4.1. *Let $D > 0$ be a non-square with $D \equiv 0, 1 \pmod{4}$. The map*

$$\iota(h, k) = (g(h, k), -k), \quad g(h, k) = \frac{D - k^2}{4h},$$

is a fixed-point-free involution of $P(D)$ which maps $P^+(D)$ onto its complement. Consequently $\#P^+(D) = \frac{1}{2}\#P(D)$.

Proof. Since $4hg = D - k^2 = D - (-k)^2$ and $g \geq 1$ is an integer dividing $(D - k^2)/4$ with complementary divisor h , the map ι sends $P(D)$ to itself, and $\iota \circ \iota = \text{id}$.

A fixed point would require $k = -k$ and $h = g$, i.e. $k = 0$ and $D = 4h^2$, contradicting that D is not a square.

By the computation in the proof of Lemma 3.1, $(h, k) \in P^+(D)$ if and only if $h + k > g$, and likewise $\iota(h, k) = (g, -k) \in P^+(D)$ if and only if $g + (-k) > h$, i.e. $g > h + k$. The remaining case $h + k = g$ would give $(2h + k)^2 = 4h^2 + 4hk + k^2 = 4hg + k^2 = D$, again forcing D to be a square. Hence exactly one of (h, k) , $\iota(h, k)$ lies in $P^+(D)$, and pairing the elements of $P(D)$ by ι gives $\#P^+(D) = \frac{1}{2}\#P(D)$. $\square$

Second proof of Theorem 1.1. Take $D = N^2 - 4$, which is positive, $\equiv 0, 1 \pmod{4}$, and not a square for $N \geq 3$. By (4) and the substitution $m = N - 2a$ of Remark 2.3,

$$\#P(D) = \sum_{\substack{k^2 < D \\ k^2 \equiv D(4)}} d\left(\frac{D - k^2}{4}\right) = \sum_{a=1}^{N-1} d(a(N - a) - 1) = \Phi(N)$$

by Proposition 2.2. By Lemma 3.1 and Lemma 4.1, $Z(D) = \#P^+(D) = \frac{1}{2}\#P(D) = \frac{1}{2}\Phi(N)$. $\square$

5. NECKLACES: THE STRUCTURAL PROOF

We now explain the factor 2.

5.1. Necklaces and conjugacy classes. By Proposition 2.1 a matrix $X \in \mathcal{M}$ is the same thing as a finite word $w(X)$ in the letters L, R . Since $\text{tr}(XY) = \text{tr}(YX)$, the trace is insensitive to the starting point of the word: writing $w = uv$ and cutting elsewhere gives $vu = u^{-1}(uv)u$, a conjugate matrix. The correct objects are therefore *necklaces*, i.e. words taken up to cyclic rotation.

Proposition 5.1. *For $N \geq 3$, the map $X \mapsto w(X)$ induces a bijection between the $\text{SL}_2(\mathbb{Z})$ -conjugacy classes of trace N and the necklaces whose matrix has trace N . If $\mathcal{C}$ is such a class and $u_{\mathcal{C}}$ is the primitive root of the corresponding necklace, then $\mathcal{C} \cap \mathcal{M}$ has exactly $|u_{\mathcal{C}}| = \#L(u_{\mathcal{C}}) + \#R(u_{\mathcal{C}})$ elements. Consequently*

$$\Phi(N) = \sum_{\mathcal{C}} (\#L(u_{\mathcal{C}}) + \#R(u_{\mathcal{C}})), \quad (6)$$

the sum being over the conjugacy classes of trace N .

Proof. Every $X \in \text{SL}_2(\mathbb{Z})$ with $\text{tr } X \geq 3$ is hyperbolic and is conjugate to an element of $\mathcal{M}$, uniquely up to cyclic rotation of its word: this is the classical theory of cutting sequences and of cycles of reduced indefinite forms, see Series [10] or Zagier [13, §13]. Rotations of a word w of primitive period p produce exactly p distinct words, hence, by freeness, exactly p distinct matrices. $\square$

5.2. The Zagier cycle counts one letter. Zagier's reduction theory [13, §13] attaches to each $\text{SL}_2(\mathbb{Z})$ -class of indefinite forms of non-square discriminant D a single cycle of Zagier-reduced forms; consecutive members satisfy $A' = C$ and $B' \equiv -B \pmod{2C}$, and the length of the cycle is the period of the *minus*, or Hirzebruch–Jung, continued fraction of the associated quadratic irrational. Forms are not assumed primitive here, in accordance with the definition of $Z(D)$; correspondingly, the classical bijection between conjugacy classes of trace N and classes of forms of discriminant $N^2 - 4$ is onto all such classes. Combining this with the conversion between regular and minus continued fractions gives the following.

Proposition 5.2. *Let $\mathcal{C}$ be a conjugacy class of trace $N \geq 3$, with primitive necklace word $u_{\mathcal{C}} = L^{a_1} R^{a_2} \dots L^{a_{2k-1}} R^{a_{2k}}$. Then the Zagier cycle of the corresponding class of forms of discriminant $N^2 - 4$ has length $a_2 + a_4 + \dots + a_{2k} = \#R(u_{\mathcal{C}})$. Consequently*

$$Z(N^2 - 4) = \sum_{\mathcal{C}} \#R(u_{\mathcal{C}}). \quad (7)$$

Proof. The regular continued fraction with purely periodic part $[a_1, a_2, a_3, \dots, a_{2k}]$ converts into the minus continued fraction with period

$$[a_1 + 1, \underbrace{2, \dots, 2}_{a_2-1}, a_3 + 2, \underbrace{2, \dots, 2}_{a_4-1}, \dots, a_{2k-1} + 2, \underbrace{2, \dots, 2}_{a_{2k}-1}].$$

Each odd-indexed partial quotient contributes one entry and each even-indexed a_{2i} contributes $a_{2i} - 1$ entries, so the minus period has length $k + \sum_i (a_{2i} - 1) = \sum_i a_{2i}$, which is the number of R's in u_C . Summing over classes and invoking Zagier's cycle theorem gives (7). $\square$

5.3. The swap. Let

$$J = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \in \mathrm{GL}_2(\mathbb{Z}), \quad \det J = -1, \quad J^2 = I.$$

Lemma 5.3. $J\mathbf{L}J = \mathbf{R}$ and $J\mathbf{R}J = \mathbf{L}$. The map $\tau(X) = JXJ$ is an involution of $\mathcal{M}$ which preserves the trace and the determinant; on words it exchanges the letters $\mathbf{L}$ and $\mathbf{R}$, hence exchanges $\#\mathbf{L}$ and $\#\mathbf{R}$; and it maps $\mathrm{SL}_2(\mathbb{Z})$ -conjugacy classes to $\mathrm{SL}_2(\mathbb{Z})$ -conjugacy classes.

Proof. The two matrix identities are immediate, and they show $\tau(\mathcal{M}) = \mathcal{M}$ with $\tau(w) = \bar{w}$, the word with its letters exchanged. Conjugation by any invertible matrix preserves the trace and determinant. Finally, if $Y = X^{-1}MX$ then $\tau(Y) = (JXJ)^{-1}\tau(M)(JXJ)$ and $JXJ \in \mathrm{SL}_2(\mathbb{Z})$ whenever $X \in \mathrm{SL}_2(\mathbb{Z})$, since $\det(JXJ) = \det X$. Thus τ carries conjugacy classes to conjugacy classes. $\square$

Remark 5.4. It is worth emphasising why this is not a triviality. J has determinant -1 , so τ is an *outer* operation: it is not conjugation by an element of $\mathrm{SL}_2(\mathbb{Z})$, and it does not fix conjugacy classes in general — indeed for $N = 4$ it exchanges the only two classes. What survives is exactly what is needed: it preserves the trace, so it permutes the finite set of classes of trace N , and it exchanges the two letter counts.

Third proof of Theorem 1.1. Fix $N \geq 3$ and let $\mathcal{S}$ be the (finite) set of conjugacy classes of trace N . By Lemma 5.3, τ induces a permutation of $\mathcal{S}$ with $\#\mathbf{L}(u_{\tau(C)}) = \#\mathbf{R}(u_C)$. Summing over $\mathcal{S}$ and reindexing by τ ,

$$\sum_{C \in \mathcal{S}} \#\mathbf{L}(u_C) = \sum_{C \in \mathcal{S}} \#\mathbf{L}(u_{\tau(C)}) = \sum_{C \in \mathcal{S}} \#\mathbf{R}(u_C).$$

Therefore, by (6) and (7),

$$\Phi(N) = \sum_C (\#\mathbf{L} + \#\mathbf{R}) = 2 \sum_C \#\mathbf{R} = 2Z(N^2 - 4). \quad \square$$

Remark 5.5. Proposition 5.2 is strictly stronger than Theorem 1.1: it is a statement about each class separately, of which the theorem is the total. The smallest instance is $N = 4$, where the two classes are $\mathbf{LLR}$ and $\mathbf{LRR}$, exchanged by τ ; their R-counts are 1 and 2, and discriminant 12 has exactly $1 + 2 = 3$ Zagier-reduced forms, while $\Phi(4) = 6$.

6. NECKLACES FIXED BY THE SWAP

For a word u write $\bar{u}$ for the word with $\mathbf{L}$ and $\mathbf{R}$ exchanged.

Proposition 6.1. *Let w be a necklace of length n whose matrix M has trace $N \geq 3$. The following are equivalent:*

- (i) w is fixed by the swap, i.e. $\bar{w}$ is a rotation of w ;
- (ii) n is even and $w = u\bar{u}$ as a necklace, for some word u of length $n/2$;

(iii) $M = V^2$ for some $V \in \mathrm{GL}_2(\mathbb{Z})$ with $\det V = -1$.

When these hold, $N = s^2 + 2$ with $s = \mathrm{tr} V \in \mathbb{Z}$. Conversely, for every $s \geq 1$ the necklace $\mathbf{L}^s \mathbf{R}^s$ is swap-invariant of trace $s^2 + 2$; hence the set of traces of swap-invariant necklaces is exactly $\{s^2 + 2 : s \geq 1\}$.

Proof. (i) $\Rightarrow$ (ii). Let σ denote rotation by one letter and suppose $\bar{w} = \sigma^j(w)$. Applying the bar operation again and using that it commutes with σ gives $w = \sigma^{2j}(w)$, so $n \mid 2j$ and $j \in \{0, n/2\}$. The case $j = 0$ would give $w = \bar{w}$ letterwise, forcing each letter to equal its own image under the swap, which is absurd. Hence n is even and $\bar{w} = \sigma^{n/2}(w)$: writing $w = uv$ with $|u| = |v| = n/2$, this reads $\bar{u}\bar{v} = vu$, whence $v = \bar{u}$.

(ii) $\Rightarrow$ (iii). If $w = u\bar{u}$ then, with U the matrix of u , the matrix of $\bar{u}$ is JUJ by Lemma 5.3, so $M = U \cdot JUJ = (UJ)(UJ) = V^2$ with $V = UJ$ and $\det V = \det U \det J = -1$.

(iii) $\Rightarrow$ (i). If $M = V^2$ with $\det V = -1$ then $JMJ = (JVJ)^2$; moreover V commutes with $M = V^2$, so M and JMJ are conjugate by $VJ \in \mathrm{SL}_2(\mathbb{Z})$ — indeed $(VJ)^{-1}M(VJ) = JV^{-1}V^2VJ = JV^2J = MJM$ — and by Proposition 5.1 conjugate elements of $\mathcal{M}$ have rotation-equivalent words, i.e. $\bar{w}$ is a rotation of w .

For the trace: $\det V = -1$ gives $V^2 = (\mathrm{tr} V)V + I$ by Cayley–Hamilton, so $N = \mathrm{tr}(V^2) = (\mathrm{tr} V)^2 + 2$. Finally $\mathbf{L}^s \mathbf{R}^s$ has matrix $\begin{pmatrix} 1+s^2 & s \\ s & 1 \end{pmatrix}$ of trace $s^2 + 2$, and equals $u\bar{u}$ with $u = \mathbf{L}^s$. $\square$

7. THE CUT-SUM OF A NECKLACE

The necklace point of view makes available an invariant that a linear word does not possess.

Definition 7.1. Let w be a necklace of length n , and let $M_0, \dots, M_{n-1} \in \mathcal{M}$ be the matrices of its n cuts. The *cut-sum* of w is

$$S(w) = \sum_{i=0}^{n-1} M_i \in M_2(\mathbb{Z}).$$

Rotating the necklace permutes the cuts, so $S(w)$ depends only on the necklace: it is a genuine matrix-valued invariant, and hence an invariant of the conjugacy class together with the choice of word length n . It refines the trace, which sees only $\mathrm{tr} S(w) = nN$.

Proposition 7.2. Let w be a necklace of length n and trace N . Then $\mathrm{tr} S(w) = nN$ and

$$\det S(w) = n + \sum_{0 \leq i < j \leq n-1} \mathrm{tr}[u_{ij}, v_{ij}], \quad (8)$$

where u_{ij} and v_{ij} are the matrices of the two arcs into which the cuts i and j divide the necklace, and $[u, v] = uvu^{-1}v^{-1}$.

Proof. The trace statement is clear since all M_i have trace N . On M_2 the determinant is a quadratic form whose polarisation is $\det(X+Y) - \det X - \det Y = \mathrm{tr} X \mathrm{tr} Y - \mathrm{tr}(XY)$; applying this repeatedly,

$$\det\left(\sum_i M_i\right) = \sum_i \det M_i + \sum_{i < j} (\mathrm{tr} M_i \mathrm{tr} M_j - \mathrm{tr}(M_i M_j)) = n + \sum_{i < j} (N^2 - \mathrm{tr}(M_i M_j)).$$

Fix $i < j$ and let $u = u_{ij}$, $v = v_{ij}$, so that $M_i = uv$ and $M_j = vu$ and $M_i M_j = uvvu$, whence $\mathrm{tr}(M_i M_j) = \mathrm{tr}(u^2 v^2)$ by cyclicity. For $X \in \mathrm{SL}_2$, Cayley–Hamilton gives $X^2 = (\mathrm{tr} X)X - I$; writing $x = \mathrm{tr} u$, $y = \mathrm{tr} v$, $z = \mathrm{tr}(uv) = N$,

$$\mathrm{tr}(u^2 v^2) = \mathrm{tr}((xu - I)(yv - I)) = xyz - x^2 - y^2 + 2.$$

Hence $N^2 - \mathrm{tr}(M_i M_j) = x^2 + y^2 + z^2 - xyz - 2$, which is $\mathrm{tr}[u, v]$ by Fricke’s identity for SL_2 . $\square$

Proposition 7.3. *Let w be a necklace of length n and trace N , put $D = N^2 - 4$, and set*

$$\Delta(w) = (\text{tr } S)^2 - 4 \det S, \quad V(w) = 2S(w) - nN I.$$

Then $V(w)$ is an integral traceless matrix and $\Delta(w) = q(V(w))$, where $q(X) = -\det X$ is the quadratic form of signature $(2, 1)$ on the traceless plane. Writing $M_i = \frac{N}{2}I + P_i$, the traceless parts P_i satisfy $q(P_i) = D/4$ and

$$\Delta(w) = 4q\left(\sum_{i=0}^{n-1} P_i\right).$$

Proof. $\text{tr } V = 2nN - 2nN = 0$ and V is integral. For traceless Y one has $\det(\alpha I + Y) = \alpha^2 + \det Y$, so with $\alpha = nN/2$ and $Y = \sum P_i$ we get $\det S = (nN/2)^2 + \det \sum P_i$, whence $\Delta = (nN)^2 - 4 \det S = -4 \det \sum P_i = 4q(\sum P_i)$. Also $V = 2 \sum P_i$, so $q(V) = 4q(\sum P_i) = \Delta$. Finally $q(P_i) = -\det(M_i - \frac{N}{2}I) = -(1 - \frac{N}{2} \cdot N + \frac{N^2}{4}) = \frac{N^2}{4} - 1 = \frac{D}{4}$. $\square$

Geometrically, P_i is the pole vector of the axis of M_i , so $\Delta(w)$ is four times the squared Minkowski length of the sum of the pole vectors of the n lifts of the closed geodesic determined by $\mathcal{C}$ — one lift for each crossing of the Farey tessellation. It is thus a centre-of-mass invariant of a closed geodesic on the modular surface.

By construction Δ is invariant under rotation, under reversal of the word, and under the swap τ ; the following was found by exhaustive computation and we state it as an observation rather than a theorem.

Observation 7.4. Let two necklaces be called *equivalent* when one is obtained from the other by rotation, reversal, and/or the swap.

- (a) For every $N \leq 34$, Δ takes distinct values on the inequivalent necklaces of trace N ; likewise for every $n \leq 14$, Δ takes distinct values on the inequivalent necklaces of length n having equal trace.
- (b) The first coincidences occur at $n = 15$, with two pairs, and at $n = 16$, with three pairs. In each of these five cases the entire matrix S agrees, not merely Δ . For instance, at $n = 15$ and $N = 576$ the inequivalent necklaces LLLRLLRLLLRLR and LLLRLLRRLRLLLRR both have $S = \begin{pmatrix} 4320 & 2984 \\ 5176 & 4320 \end{pmatrix}$.

We have no explanation for these coincidences; see §10.

8. A TOTAL-MATRIX IDENTITY

Proposition 2.2 evaluates not only the cardinality of a trace fibre but any sum of matrix entries over it.

Theorem 8.1. *For every $N \geq 3$, with $J = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$,*

$$\sum_{\substack{X \in \mathcal{M} \\ \text{tr } X = N}} X = \frac{N}{2} \Phi(N) \cdot I + \left(\sum_{a=1}^{N-1} \sigma(a(N-a)-1) \right) \cdot J.$$

In particular $N\Phi(N)$ is even, which also follows from Theorem 1.1.

Proof. By Proposition 2.2 the fibre is parametrised by pairs (a, b) with $1 \leq a \leq N-1$ and $b \mid \mathcal{N}_a = a(N-a)-1$, the matrix being $\begin{pmatrix} a & b \\ \mathcal{N}_a/b & N-a \end{pmatrix}$. Summing the $(1, 1)$ entries gives $\sum_a a d(\mathcal{N}_a)$; since $\mathcal{N}_a = \mathcal{N}_{N-a}$, replacing a by $N-a$ shows this equals $\sum_a (N-a) d(\mathcal{N}_a)$, so twice it is $N \sum_a d(\mathcal{N}_a) = N\Phi(N)$ by (3). The $(2, 2)$ entries give the same value. Summing the

(1, 2) entries gives $\sum_a \sum_{b|\mathcal{N}_a} b = \sum_a \sigma(\mathcal{N}_a)$, and the (2, 1) entries give $\sum_a \sum_{b|\mathcal{N}_a} \mathcal{N}_a/b$, which is the same because $b \mapsto \mathcal{N}_a/b$ permutes the divisors of $\mathcal{N}_a$. $\square$

Remark 8.2. The diagonal of Theorem 8.1 is Φ itself, the object of [5, 4, 1, 12, 11], whose analysis rests on the divisor sums of Hooley [3] and Bykovskii–Ustinov [2]. The off-diagonal entry $\sum_a \sigma(a(N-a)-1)$ is the corresponding sum-of-divisors problem for the same quadratic polynomial. We are not aware of any treatment of it, and its asymptotic behaviour — in particular, whether it is as irregular as Φ — appears to be open.

9. COMPUTATIONAL VERIFICATION

All computations were carried out in exact integer arithmetic; no floating-point value enters any decision (inequalities such as $2h+k > \sqrt{D}$ are tested as $2h+k > 0$ together with $(2h+k)^2 > D$). Both sides of each identity were computed by independent routines: the left-hand sides by enumerating the free monoid or the necklaces directly, the right-hand sides by enumerating triples (A, B, C) subject to (2) and, separately, by the divisor parametrisation of Lemma 3.1; the resulting values of Φ and Z were checked against the terms published in A264598 and A257007.

Assertion	Range verified	Outcome
$\Phi(N) = \sum_a d(a(N-a)-1) = \#P(N^2-4) = 2Z(N^2-4)$	$N = 3, \dots, 150$	148/148
$\sum_c \#L = \sum_c \#R = Z(N^2-4)$	$N = 3, \dots, 150$	148/148
ι fixed-point-free, condition-reversing, every pair	$N = 3, \dots, 90$	no exception
Per class: Zagier cycle lengths = $\#R$ as multisets	$N = 3, \dots, 24$	identical
$\det S = n + \sum \text{tr}[u, v]$ and $\Delta = q(2S - nNI)$	all necklaces, $n \leq 9$	0 failures
Δ separates inequivalent necklaces	$N \leq 34; n \leq 14$	no collision
First Δ -collisions	$n = 15, 16$	2, then 3 pairs
Theorem 8.1	$N = 3, \dots, 90$	88/88

The first ten values are as follows.

N	3	4	5	6	7	8	9	10	11	12
$\Phi(N)$	2	6	8	14	14	24	16	40	26	36
$Z(N^2-4)$	1	3	4	7	7	12	8	20	13	18
$\sum_a \sigma(a(N-a)-1)$	2	10	20	45	62	120	120	270	250	380

10. OPEN PROBLEMS

- (1) Explain Observation 7.4(b). The five known coincidences are presumably the beginning of an infinite family; is there a construction producing pairs of inequivalent necklaces with equal cut-sum? Equivalently: which pairs of closed geodesics of the same length have the same centre of mass over their Farey crossings?
- (2) Determine the asymptotic behaviour of $\sum_{a=1}^{N-1} \sigma(a(N-a)-1)$, the off-diagonal entry of Theorem 8.1. Does it admit a limiting distribution after normalisation, as Φ does by Peter’s theorem [9]?
- (3) Is Δ , or the full cut-sum S , expressible in terms of classical invariants of the corresponding ideal class? Proposition 7.3 shows Δ is the value at an explicit lattice vector of the ternary form q ; the configuration of the $h(N^2-4)$ vectors $V(w)$ attached to a single discriminant deserves study.
- (4) The OEIS entries A264597, A264598 and A257007 should have their conjecture annotations replaced by a reference.

DATA AND CODE AVAILABILITY

The verification programs are plain Python 3 using only the standard library. Running `final.py` performs the end-to-end check of every identity above and reports either **ALL CLAIMS VERIFIED** or the first counterexample.

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